Distinct Proteoform Profiles in Adult Zebrafish Telencephalon and Optic Tectum

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Abstract

Adult zebrafish telencephalon and optic tectum differ in function, yet it has remained unclear how clearly that distinction appears at the proteoform level. We addressed that question using the published article-level counts together with the released Tel2/Teo2 tables from a 2022 region-resolved top-down proteomics study. The two regions proved to be strongly separated rather than mildly shifted versions of the same local proteome. Only 35 proteoforms were shared across a union of 807 in the published study, and the same low-overlap pattern remained in the released exact-ID tables, after protein collapse, and after conservative tests for uneven detectability and identifier error. The separation was also biologically coherent. Telencephalon-associated markers of neuropeptide and synaptic biology concentrated in telencephalon, whereas optic-tectum-associated structural and reticulon-linked proteoforms concentrated in optic tectum. An independent literature-derived panel built from adult zebrafish myelin and optic-tectum radial-glia markers supported the same regional split. A possible excess of N-terminally acetylated proteoforms in telencephalon became much less convincing once we accounted for unequal recovery between the two regions, so we treat that pattern as suggestive rather than settled. These results show that the released dataset already supports a clear biological conclusion: adult zebrafish telencephalon and optic tectum have distinct proteoform profiles, and the proteoforms that distinguish them follow the known biology of the two regions.

1 Introduction

Region-resolved proteomics is most informative when it can expose molecular differences that parallel functional anatomy. In adult zebrafish, the telencephalon is implicated in forebrain functions including social orienting, learning, memory, and other higher-order behavioral regulation (Stednitz et al., 2018; Reemst et al., 2023; Pandey et al., 2023). The optic tectum, by contrast, is a classic visuomotor structure that transforms sensory information into orienting and avoidance actions (Roeser and Baier, 2003; Orger, 2016). If those regions maintain distinct proteoform repertoires, a region-resolved top-down proteomics record should be able to detect that difference.

The 2022 LCM-CZE-MS/MS study of Lubeckyj and Sun (2022) offers a useful test. It isolated adult zebrafish telencephalon and optic tectum and identified hundreds of proteoforms from each region using a top-down workflow whose downstream interpretation would normally depend on tools such as TopPIC and related proteoform-level post-processing (Choi and Liu, 2022). The article argued that the resulting profiles matched basic regional function, but the evidence for that interpretation remained dispersed across the text and figures. The accompanying PRIDE release exposes region-specific Tel2 and Teo2 proteoform tables, making it possible to examine exact accession+proteoform overlaps, duplicate incidence patterns, marker-family membership, and simple intensity summaries without pretending to have rerun the full raw-feature pipeline.

The question addressed here is therefore straightforward: what does this released top-down dataset already establish about regional molecular organization in the adult zebrafish brain? Read quantitatively, it supports a sharply separated telencephalon-versus-tectum proteoform architecture that also tracks known regional biology. Just as important, the released duplicate structure lets us ask whether that separation could be explained away by uneven recovery or identifier noise. Within the limits of the public tables, it cannot. Prior zebrafish functional studies and atlases tell us what the two regions are generally for (Stednitz et al., 2018; Pandey et al., 2023; Roeser and Baier, 2003; Orger, 2016), while recent top-down zebrafish-brain studies show that proteoform-level structure remains biologically informative well beyond this original pilot (Askarani et al., 2025). In parallel, spatial-omics foundation models such as KRONOS and HEIST emphasize large-scale representation learning across imaging or count-based spatial assays (Shaban et al., 2025; Madhu et al., 2025). What has been missing is a compact proteoform-specific account that states, in quantitative terms, what this public dataset already implies about regional molecular organization.

Three results follow. First, the released telencephalon and optic-tectum tables imply a much sharper proteoform-level separation than a casual reading of the source paper would suggest, and that result is visible across article-level, exact-ID, canonicalized, protein-collapsed, prevalence-adjusted, and duplicate-based detectability views. Second, the separation is biologically ordered rather than merely sparse: once the source paper's marker prose is rewritten as a matched-versus-spillover problem, the signal remains strong under exact testing, interval estimation, family-size-preserving randomization, protein collapse, and an independent literature-derived composition panel that is not inherited from the source paper. Third, the initially suggestive acetylation difference becomes much weaker once unequal recovery between the two regions is taken seriously, so it is better treated as a lead for future work than as an established regional program. Throughout, the analysis is packaged with code, figures, exact marker-family membership tables, standards-aware canonicalization examples, and machine-readable evidence traceability so that the argument can be checked locally rather than taken on trust.

2 Data and Analysis Setup

2.1 Source material

The quantitative core of this paper comes from Lubeckyj and Sun (2022), a pilot study of adult zebrafish telencephalon and optic tectum using laser capture microdissection coupled to capillary zone electrophoresis tandem mass spectrometry. We use three public source layers: the article itself, the journal supplementary workbook, and the released PRIDE region tables `Proteoform_ID_Tel2.xlsx` and `proteoform_ID_Teo2.xlsx`. The supplement notes that the released proteoforms were identified in TopPIC with 1% proteoform-spectrum-match FDR and 5% proteoform-level FDR, with duplicate label-free quantification performed in TopDiff. The `Tel2` and `Teo2` sheets in the supplement match the standalone PRIDE tables exactly, so we use the workbook as provenance metadata and the PRIDE tables as the quantitative source. We parse the released tables directly, preserve row-level provenance in local CSV and JSON traceability files, and record the public raw-file links in the manifest. We do not attempt raw-file reprocessing here because the processed feature histories needed for a faithful rerun are not part of the public release.

For exact-ID overlap we treat the biological unit as the accession-plus-proteoform pair. This is stricter than comparing only proteoform strings, because the same peptide-like string can occur under more than one accession in the released tables. For the curated marker panel, we use transparent matching rules tied to the source paper's own language: exact gene-symbol matches for NPY, PENKB, PYY-A, and neurogranin a, and description-keyword matches for synucleins, reticulon, myosin, actin, and troponin. Those rules are written into the local traceability files so that a reader can inspect or contest them directly.

2.2 Derived metrics

The computation asks three related questions. First, how strongly are telencephalon and optic tectum separated when the released proteoforms are compared at exact-ID, canonicalized, protein-collapsed, and abundance-weighted levels? Second, do the proteoforms highlighted by the biological literature and by the source paper concentrate in the expected region? Third, does the visible PTM layer add a stable regional signal or merely a suggestive one?

To answer the first question, we compare the article-level overlap, the exact-ID overlap from the released source tables, a canonicalized discrepancy diagnostic, abundance-weighted similarities, a prevalence-adjusted exact overlap test, a duplicate-based detectability correction, and conservative misidentification bounds keyed to the 1% PrSM and 5% proteoform-level FDR values reported for the source workflow. To answer the second, we score both the source-paper marker panel and an independent literature-derived composition panel by asking how often the named markers fall in their expected region. To answer the third, we examine the N-terminal acetylation contrast in the matched marker subsets and then test whether that contrast survives simple sensitivity analyses for unequal recovery.

We also run a compact set of reviewer-motivated robustness checks that the public release can genuinely support: alternative abundance normalizations, duplicate-informed missingness bounds, a minimal MNAR-style low-intensity fill, family-size-preserving randomization, conservative spillover reassignment, motor-family exclusion, protein collapse, and an independent panel built from adult zebrafish myelin and optic-tectum radial-glia markers. The formulas, normalization definitions, canonicalization rules, detectability model, and missingness calculations are given in the Appendix so that the main text can stay close to the biological question.

2.3 Interpretive scope

The analysis stays close to what the public release can genuinely support. It does not recover the raw TopPIC feature tables, estimate proteome-wide significance for every detected proteoform, or reinterpret unidentified mass shifts. Its strongest claims therefore remain about regional separation and about a deliberately interpretable marker layer. What the released region tables do provide, however, is enough structure to move from narrative suggestion to an explicit statement about adult zebrafish regional proteoform organization, and to show that this statement does not disappear under simple detectability or identifier-error bounds.

3 Results

3.1 Telencephalon and optic tectum are sharply separated at the proteoform level

The first substantive result is a new quantitative statement about adult zebrafish brain organization: the released top-down dataset shows that telencephalon and optic tectum have sharply distinct proteoform profiles. At the article level, telencephalon contributes 309 identified proteoforms and optic tectum contributes 533, but only 35 proteoforms are shared, yielding a Jaccard overlap of 0.0434. The released source tables tell the same story while making the evidence chain tighter. Exact accession+proteoform matching across `Proteoform_ID_Tel2.xlsx` and `proteoform_ID_Teo2.xlsx` yields 29 shared entries over a union of 805, corresponding to an exact-ID Jaccard of 0.0360 and an exact-ID Sørensen overlap of 0.0689. Using mean duplicate intensities barely changes the picture: the weighted Jaccard is 0.0427 and the Bray–Curtis similarity is 0.0820. Even at the protein level, overlap remains modest at 0.2151. A prevalence-adjusted lower-tail overlap test sharpens the same point: relative to the fixed-margin independence baseline, the exact-ID centered Jaccard is -0.2792 with $p = 5.56 \times 10^{-183}$, and the protein-collapsed overlap is still far below expectation ($p = 5.84 \times 10^{-33}$). The released dataset therefore does not describe two lightly shifted samples. It describes two strongly separated regional proteoform profiles.

The discrepancy between the article's 35 shared proteoforms and the table-derived exact-ID count of 29 is also understandable rather than alarming. Relaxing the matching rule by stripping bracketed PTM annotations and punctuation while keeping accessions fixed raises the shared count to 44, placing the article

Table 1: Core regional separation metrics supporting the regional-organization claim. The first two rows preserve the article-level summary, and the remaining rows use the released exact-ID tables plus reviewer-driven detectability and identifier-error bounds.

Metric	Value
Article shared proteoforms	35
Article Jaccard overlap	0.0434
Exact-ID shared proteoforms	29
Exact-ID Jaccard overlap	0.0360
Weighted Jaccard overlap	0.0427
Protein overlap fraction	0.2151
Occupancy-adjusted shared exact IDs	43.2
Occupancy-adjusted Jaccard overlap	0.0432
5% misidentification ceiling	0.0540

value between strict and relaxed deterministic encodings. The practical conclusion is unchanged across those choices: none of them turns this into a high-overlap dataset.

The duplicate-resolved tables let us test whether undersampling could still hide a largely shared regional proteome. They do not support that interpretation. Chao2 and jackknife lower-bound analyses keep the overlap in the same low range, and a minimal MNAR-style low-intensity fill changes the abundance-weighted similarities only slightly. Across the alternative normalizations and sensitivity checks, the overlap remains low enough that the main biological picture does not move.

Depth differences explain only a small part of the split. In a simple two-run occupancy-style model built from duplicate rediscovery, the implied per-run detection probability is 0.665 in telencephalon and 0.506 in optic tectum. Correcting the observed shared exact IDs for those effort differences raises the implied shared count from 29 to 43.2 and the Jaccard overlap only to 0.0432. That adjusted value is still close to the observed proteoform overlap and far below the protein-level overlap, so uneven recovery sharpens the split but does not create it.

The same conclusion holds under conservative identifier-error bounds. If 5% of region-unique exact IDs are treated as possible proteoform-level false discoveries and collapsed in the direction most favorable to overlap, the Jaccard ceiling rises only to 0.0540. Even a 10% bound reaches just 0.0725. Uneven detectability and plausible identifier error can soften the exact edge of the split, but neither turns the two regions into variants of the same local proteoform pool.

3.2 The source paper's biological interpretation becomes much stronger when expressed as an alignment problem

A second substantive result is that this regional split is biologically organized in the expected direction. The curated marker panel is highly concentrated in the expected region. Telencephalon-associated markers contribute 23 matched counts and only 2 spillover counts, whereas optic-tectum-associated markers contribute 102 matched counts and 1 spillover count. Across the whole panel, the observed matched-region fraction is 0.9766. A baseline built only from the panel's own 25-versus-103 telencephalon/optic-tectum prevalence would predict 0.6904, so the observed alignment exceeds that baseline by 0.2861. This matters because optic tectum contributes the larger raw proteoform total in the source study; the alignment result is not just a restatement of optic-tectum abundance.

Written as a 2×2 table, the marker counts yield an odds ratio of 1173 with an approximate 95% confidence interval from 102.0 to 13,495.1 and a one-sided exact p value of 5.12×10^{-22}. The family-size-

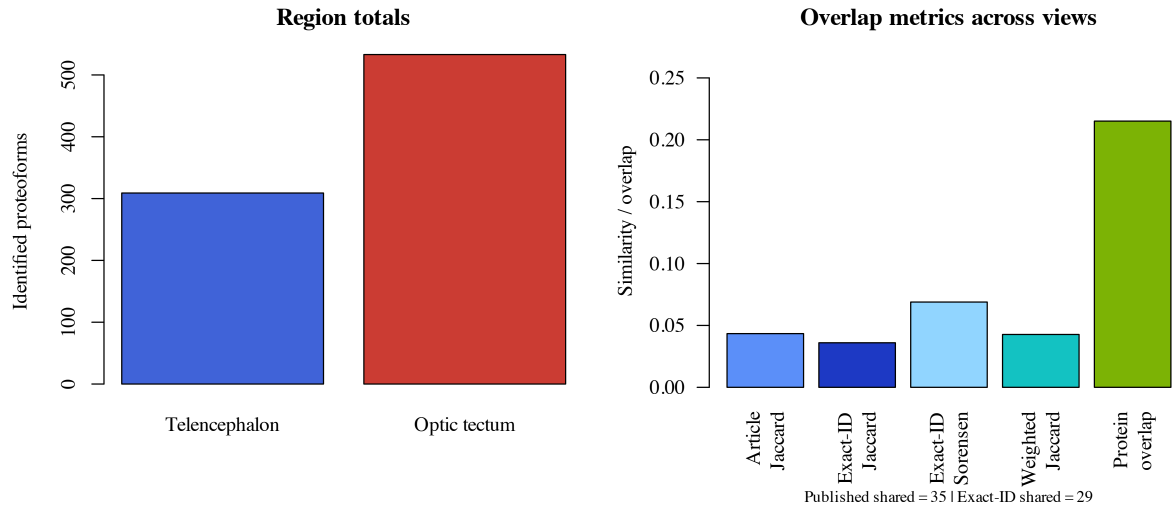

Figure 1: Regional totals and overlap metrics. The released tables contain 303 exact IDs in telencephalon and 531 in optic tectum, with 29 exact-ID overlaps. The exact-ID source-table view is slightly stricter than the article aggregate, but both support the same low-overlap picture.

preserving randomization makes the same point in a way that does not depend on asymptotic odds-ratio logic: under 200,000 randomizations that preserved the nine family sizes and the global 25-versus-103 telencephalon/optic-tectum label totals, none reached the observed 125 matched counts. The count-level result also survives stronger robustness checks. All 9 curated marker groups favor their expected region, collapsing the panel to unique proteins still yields a strong regional signal, and intensity weighting does not weaken the alignment. These results do not classify the whole proteome. They show that the proteoforms the source study itself treated as biologically meaningful are overwhelmingly concentrated in the region whose known function they should support.

The stress tests point in the same direction. Leave-one-out removal of any single marker family keeps the matched-region fraction between 0.9625 and 0.9911. Conservative spillover reassignment still leaves the panel strongly aligned, and removing the motor-associated myosin, actin, and troponin families leaves a remaining alignment fraction of 0.9310. A lightweight composition guardrail also argues against a simple contamination explanation: the released tables show only 2 optic-tectum glial-support sentinels but 23 telencephalon myelin sentinels versus 3 in optic tectum.

An independently curated panel that was not inherited from the source paper's own prose reaches the same qualitative result. Using adult zebrafish myelin markers from the zebrafish CNS myelin literature together with adult optic-tectum radial-glia markers from the tectal precursor literature yields 32 matched counts and only 3 spillover counts, for an alignment fraction of 0.9143 and a one-sided exact p value of 6.68×10^{-4}. The panel is smaller and more composition-oriented than the source paper's functional families, so it should not replace them. What it does show is that the regional separation signal is broader than one interpretive choice.

3.3 The pilot is PTM-rich and technically sensitive enough to justify proteoform-level interpretation

The PTM follow-up contributes a different kind of result. It shows where the public evidence stops being firm. The source article reports 807 identified proteoforms in total, of which 358 carry mass shifts and 211 are N-terminally acetylated. That alone reinforces an important point: the dataset is not just a protein list in a

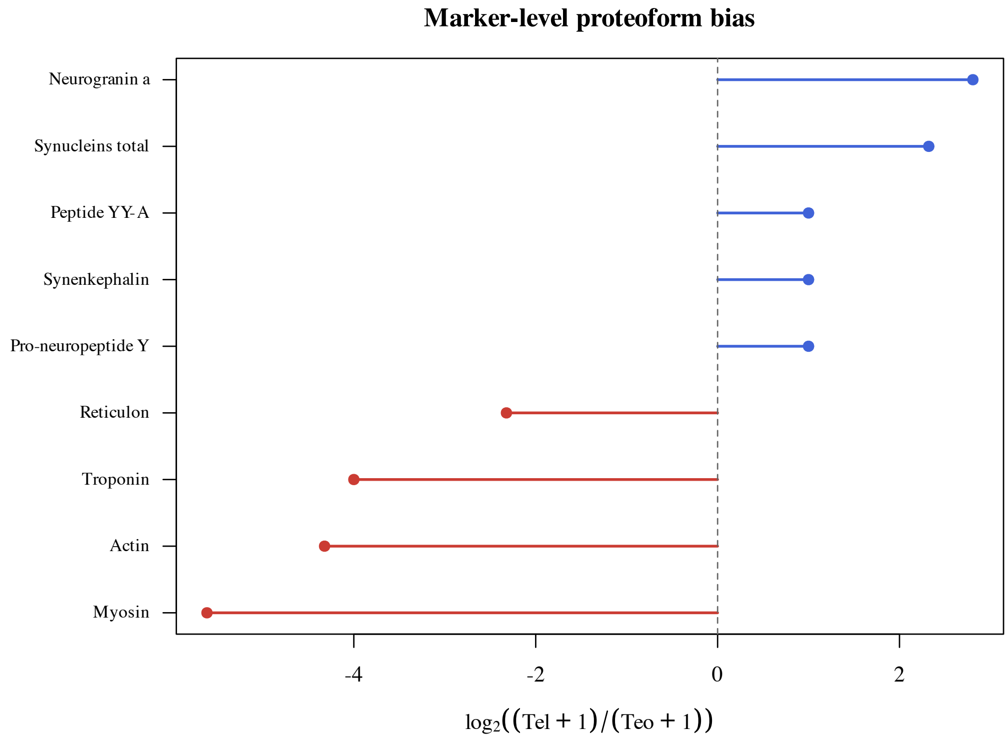

Figure 2: Marker-level proteoform bias across 128 curated proteoform observations in nine marker families. Positive values favor telencephalon and negative values favor optic tectum.

different format. A large part of the signal lives at the proteoform level.

The source tables let us connect that PTM layer back to the regional marker analysis, but the reviewer-driven follow-up changes how strongly we should read it. In the raw matched-marker counts, 14 of 23 telencephalon-matched proteoforms are N-terminally acetylated, compared with 16 of 102 optic-tectum-matched proteoforms. That looks interesting at first pass. Once precursor mass, sequence span, first-residue position, and unequal recovery between the two regions are taken seriously, however, the effect weakens markedly. The adjusted odds ratio falls to 2.94 (95% CI 0.70–12.30; $p = 0.140$), and the first-residue≤ 2 subset yields no enrichment. The right reading is therefore cautious: the released tables suggest a possible regional difference in acetylation, but they do not establish a robust regional PTM program.

The duplicate-sensitivity values help explain why this pilot can still support a clear biological result. The duplicate analyses yielded means of 232 proteoforms for telencephalon and 356 for optic tectum, corresponding to recovery fractions of 0.7508 and 0.6680 relative to the regional totals. The single-run high-water mark was 418 proteoforms from an injected amount corresponding to roughly 250 cells in the source paper. That does not remove the pilot-study caveat, but it helps explain why a biologically legible regional structure is already visible in a very small design.

4 Discussion

The central result of this study is a proteoform-resolved regional separation between adult zebrafish telencephalon and optic tectum. The original study made that outcome plausible. The quantitative evidence assembled here makes it explicit. Once the named markers are rewritten as a matched-versus-spillover

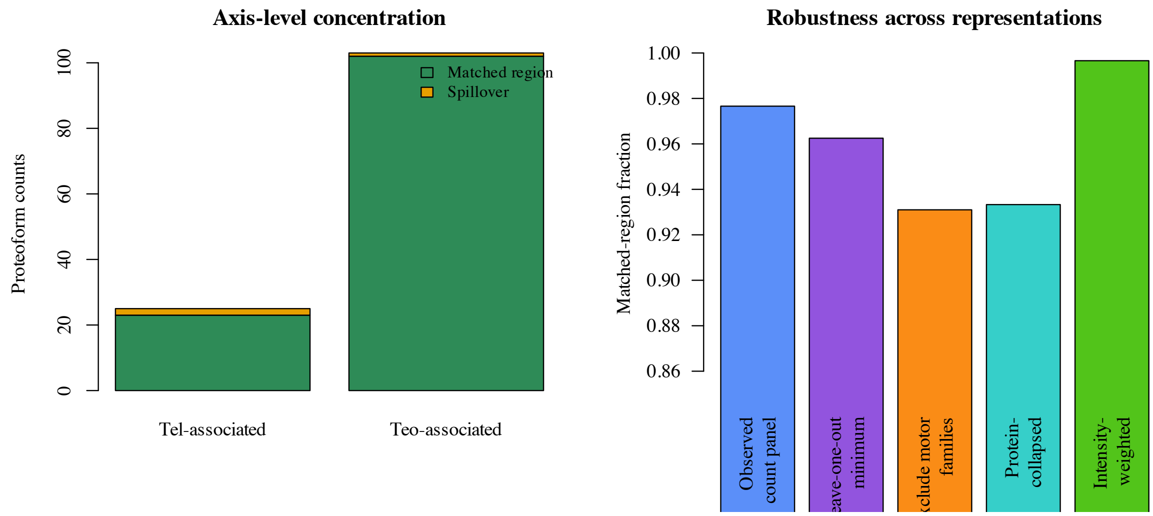

Figure 3: Axis-level concentration and robustness summaries for the nine curated marker families. The overall alignment signal remains high after leave-one-out perturbation, after excluding motor families, after collapsing to proteins, and after weighting by mean duplicate intensity.

problem, the regional story is no longer a handful of appealing examples. The alignment fractions, prevalence-adjusted overlap tests, independent panel, and stress tests all point in the same direction.

Biologically, that separation makes sense. The telencephalon is a forebrain territory involved in social orienting, learning, and memory, whereas the optic tectum is a visually driven sensorimotor structure. The marker architecture seen here follows that division of labor. Telencephalon-associated neuropeptide and synaptic markers such as NPY, PENKB, PYY-A, neurogranin a, and synuclein-family entries concentrate where a forebrain integrative role would lead one to expect them. The optic-tectum side is dominated by reticulon, myosin, actin, and troponin-associated proteoforms, consistent with a region built around visually guided orienting and the underlying cytoskeletal and projection architecture that supports it. The interesting point is not just that the two regions look different. It is that the proteoform differences remain interpretable in light of known regional biology.

The strongest conceptual gain over the source paper is therefore not a new assay. It is a clearer view of what the released top-down data already contain. Proteoform-resolved overlap is far lower than protein-level overlap, and the marker families most legible in the public release are not scattered randomly across the two regions. They follow a region-function axis. That is the sense in which this paper adds something new to the knowledge base: it turns a suggestive pilot result into a sharper statement about regional molecular organization.

The independent composition panel is useful in the same way. Myelin markers remain associated with telencephalon, and optic-tectum radial-glia markers remain associated with optic tectum. That second panel is small, but it matters because it shows that the result is not confined to one set of favored examples. The same regional contrast survives in a broader biological vocabulary.

The released Tel2 and Teo2 tables also matter because they expose the proteoform layer more faithfully than the article narrative alone. They let us compare strict accession+proteoform overlap, relaxed discrepancy diagnostics, duplicate incidence, and marker-family membership from machine-readable sources. That extra layer reveals a small provenance wrinkle: the article reports 35 shared proteoforms, whereas the released exact-ID tables expose 29 exact matches. The relaxed matching diagnostics suggest that this is better understood as a representation issue than as a biological contradiction. What matters biologically is that every public view still lands on the same conclusion: very low proteoform overlap and much higher protein overlap.

Table 2: Curated marker families used in the axis-alignment analysis. The released source tables now expose exact accession+proteoform members for each family in the local marker-membership file, but the biological framing still follows the source paper's named marker groups.

Marker group	Tel	Teo	Interpretation
Pro-neuropeptide Y	1	0	feeding-related neuropeptide signaling
Synenkephalin	1	0	affective signaling
Peptide YY-A	1	0	peptide hormone signaling
Neurogranin a	6	0	memory-linked synaptic signaling
Synucleins total	14	2	synaptic and mnemonic family
Reticulon	0	4	retinotectal organization
Myosin	0	48	motor machinery
Actin	0	19	cytoskeletal movement
Troponin	1	31	contraction-linked regulation

Table 3: Axis-level alignment compared with the naive baseline implied by regional prevalence.

Functional axis	Observed	Wilson 95% CI	Baseline	Lift
Telencephalon-associated	0.9200	0.7503–0.9778	0.1875	0.7325
Optic-tectum-associated	0.9903	0.9470–0.9983	0.8125	0.1778

The new detectability and identifier-error bounds sharpen that point. A simple duplicate-based occupancy correction raises the exact-ID Jaccard only from 0.0360 to 0.0432, even though it allows for unseen shared proteoforms hidden by uneven regional recovery. A conservative 5% false-discovery bound, chosen to match the proteoform-level FDR reported in the supplement, raises the Jaccard ceiling only to 0.0540. These are not trivial corrections, but they do not change the basic reading of the data. The two regions remain much farther apart at the proteoform level than at the protein level.

The sensitivity work is important because it tells us what would have to be true to erase the signal. The answer is: quite a lot. Conservative spillover reassignment weakens the alignment but does not remove it. Excluding the largest motor-associated families still leaves a strong regional signal. Duplicate-informed missingness bounds, bootstrap resampling, and minimal MNAR-style fills never turn the dataset into a high-overlap case. The paper still rests on public tables rather than raw-feature histories, but the regional result does not depend on a single fragile choice.

Standards matter here as well. The discrepancy between strict and relaxed matching is small enough that the biology stays the same, but it shows how easily proteoform comparison can be distorted by notation rather than by substance. ProForma established a common proteoform language, and ProForma 2.0 pushes that standard further by separating sequence, modification tokens, and positional information in a machine-readable form (LeDuc et al., 2018, 2022). The public tables do not contain enough localization metadata for full ProForma 2.0 serialization of every entry, but the bridge file in this package now breaks the source strings into those components so that the canonicalization step can be inspected in community-standard terms.

The PTM follow-up contributes in a different way. It marks the edge of what the public data can support. The raw counts make acetylation look regionally asymmetric, but that difference becomes much less convincing once detectability proxies and unequal recovery are taken seriously. That is a valuable result in itself. It keeps a potentially attractive story from outrunning the evidence and shows that this dataset is strongest when read as evidence for regional proteoform separation rather than for a settled regional PTM program.

Table 4: Sensitivity checks requested by the external review. The motor-family exclusion uses a Haldane correction because one spillover cell is zero after exclusion.

Scenario	Alignment	Odds ratio	One-sided p
Observed panel	0.9766	1173.0	5.12×10^{-22}
Shift 1 matched count per axis	0.9609	370.3	2.01×10^{-19}
Shift 2 matched counts per axis	0.9453	175.0	3.73×10^{-17}
Shift 3 matched counts per axis	0.9297	99.0	3.93×10^{-15}
Exclude myosin, actin, troponin	0.9310	84.6	6.32×10^{-4}

Table 5: Technical sensitivity values reported in the source study and extended with exact-ID duplicate-incidence counts from the released source tables.

Region	Mean	SD	CV	Recovery	exact-ID q_1	Chao2
Telencephalon	232	28	0.1207	0.7508	152	379.5
Optic tectum	356	88	0.2472	0.6680	351	873.2

4.1 Limits of inference

This is still a pilot-scale argument drawn from released tables rather than from a new experiment or a raw-file reprocessing workflow. Its strongest claims remain about regional separation and about a deliberately interpretable marker layer, not about every proteoform in the adult zebrafish brain. The released duplicate tables are enough for missingness bounds, normalization checks, explicit marker traceability, a simple occupancy-style detectability correction, and a limited PTM screen, but they do not expose the full feature-level histories, localization confidence, or cross-region replicate structures that a raw TopPIC export would provide (Choi and Liu, 2022). The current package therefore cannot answer every redundancy or detectability objection. It can show that the obvious objections do not overturn the regional-function signal.

Those limits also point to the next useful work. The clearest extension is to combine the public raw files with processed search intermediates so that the same regional question can be asked at feature level and under richer missingness models (Karpievitch et al., 2012; Lazar et al., 2016). A second is to test whether the same alignment framing remains informative across additional brain regions. A third is to make the purity question more direct rather than inferential. Even in its current form, however, the paper makes one point clearly: public top-down proteomics data can already reveal a biologically coherent regional architecture in the adult zebrafish brain when the proteoform layer is made explicit.

5 Conclusion

Taken together, these results support a proteoform-level distinction between adult zebrafish telencephalon and optic tectum that is both quantitative and biologically coherent. That conclusion survives prevalence-adjusted overlap tests, duplicate-informed Chao2 and jackknife sensitivity analyses, a simple occupancy-style detectability correction, conservative identifier-error bounds, MNAR-style low-intensity fills, family-size-preserving randomization, protein collapse, and an independent literature-derived composition panel. The PTM follow-up points in a more cautious direction: once public detectability proxies are taken seriously, the acetylation asymmetry is better read as hypothesis-generating than as a robust regional program. The evidence base remains public and limited, but it is already strong enough to support a clear statement about regional molecular organization in the adult zebrafish brain.

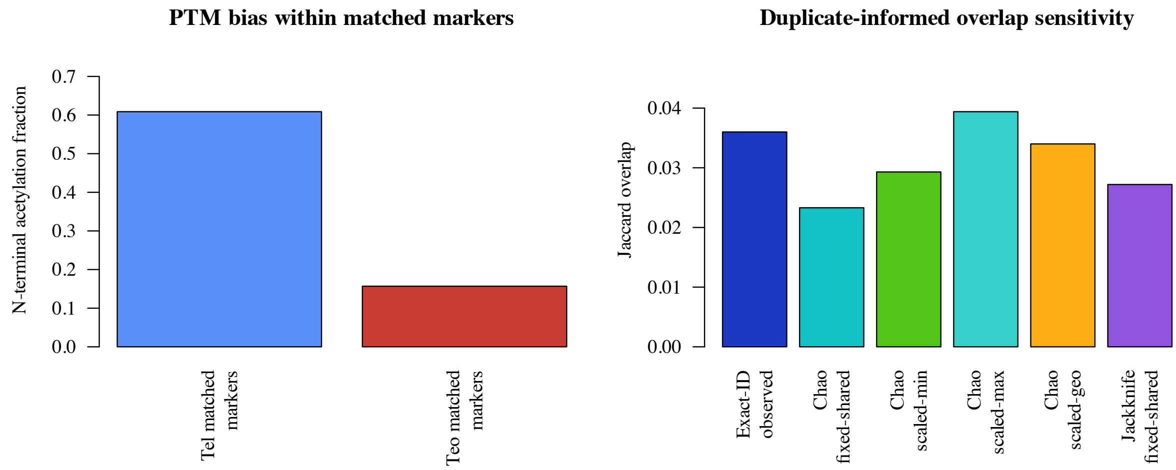

Figure 4: PTM and missingness-sensitive follow-up analyses. Left: N-terminal acetylation in the matched-marker subsets (23 telencephalon counts and 102 optic-tectum counts). Right: duplicate-informed Chao2 sensitivity changes the exact-ID Jaccard modestly, but it never turns the regional split into a high-overlap pattern.

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A Reproducibility Appendix

A.1 Local files

The package is backed by the source files in `data/`, the local analysis and figure scripts in `results/`, the build scripts in `code/`, and the environment metadata in `code/environment_versions.json`.

A.2 Formulas

The main derived formulas are the Jaccard overlap, specialization fractions, marker-level stabilized log-bias, Wilson intervals for axis-level alignment fractions, an approximate odds-ratio interval for the 2×2 matched-versus-spillover table, an exact test on the same table, a prevalence-adjusted overlap-significance check based on the fixed-margin hypergeometric null, a two-run detectability correction based on duplicate rediscovery, and a conservative misidentification ceiling based on paired collapse of region-unique IDs.

A.3 Extended computational details

The main text reports only the quantities needed to follow the biology. The full computational path remains local and explicit. Exact-ID overlap is defined on accession-plus-proteoform pairs. The discrepancy diagnostic compares that strict unit with relaxed accession-plus-canonicalized-sequence matches that strip bracketed PTM annotations and punctuation while preserving accession identity. Protein-collapsed summaries merge rows at the protein identifier level. Abundance-weighted comparisons use mean duplicate intensities within each region and report both weighted Jaccard and Bray–Curtis similarity.

For prevalence-adjusted overlap, the observed telencephalon and optic-tectum margins are held fixed and the overlap count is evaluated against the fixed-margin hypergeometric null. The paper reports the observed Jaccard, the marginal-prevalence baseline under independence, the centered Jaccard, and the exact lower-tail p value. For marker alignment, the analysis uses a matched-versus-spillover table together with Wilson intervals, an approximate odds-ratio interval, and an exact directional test. Family-size-preserving randomization keeps family sizes and the global telencephalon/optic-tectum label totals fixed while shuffling labels across the panel counts.

Missingness and abundance sensitivities are likewise kept local and inspectable. The duplicate `Match`/`No match` flags in the released tables provide the incidence counts used for Chao2 and first-order jackknife lower-bound richness calculations. Blank intensity cells are treated as absent detections in the baseline analysis, and the MNAR-style sensitivity fills missing exact IDs to half of the region-specific 5th-percentile or 10th-percentile nonzero intensity. Alternative abundance normalizations include per-run total-intensity scaling before duplicate averaging, region-level total-sum scaling, upper-quartile scaling, and median-ratio scaling on shared exact IDs.

The detectability bound uses the same duplicate structure in a different way. After collapsing each region to unique accession-plus-proteoform IDs, each exact ID is treated as detected in one run or both runs. Under a simple homogeneous two-run model, the duplicate-rediscovery counts give a region-specific per-run detection probability and therefore an estimated "seen at least once" probability for that region. The observed cross-region shared exact IDs are then divided by the product of those two region-level detection probabilities to estimate how much overlap could be hidden by uneven duplicate recovery alone. This is intentionally simpler than a full hierarchical model, but it answers the reviewer's immediate question with quantities that are actually supported by the released duplicate tables.

The misidentification sensitivity files are equally conservative. They do not claim to estimate the true false-discovery process. Instead, they ask how much the exact-ID Jaccard could rise if an equal fraction of region-unique exact IDs in both regions were actually paired representation or identification errors. The 1% and 5% scenarios track the PrSM- and proteoform-level FDR values reported in the source supplement. A second set of rows asks what error fractions would be required to raise the exact-ID Jaccard to 0.10 or all the way to the observed protein-level overlap.

A.4 Scope note

The computation uses values recoverable from the published article and the released region tables. It does not claim a full raw-data reprocessing pipeline.

A.5 Canonicalization and missing-intensity note

Canonicalization is used only as a discrepancy diagnostic, not as the paper's primary biological unit. The strict analysis unit remains accession plus proteoform string. In the diagnostic view, bracketed PTM annotations, punctuation, and non-sequence characters are removed before matching. The file `results/canonicalization_examples.csv` now separates each example into sequence core, modification tokens, residue window, and canonicalized sequence so that the discrepancy can be inspected in a ProForma-oriented way (LeDuc et al., 2018, 2022). We do not emit full ProForma 2.0 strings for the whole dataset because the public tables do not expose the localization metadata needed for exact standard-compliant serialization. Blank intensity cells in the released tables are treated as absent detections in the baseline analysis and contribute zero intensity; the reviewer-driven MNAR sensitivity file records two low-intensity-limit alternatives.

A.6 Reproducibility metadata

The package records software versions and platform metadata in `code/environment_versions.json`. It also records the public repository URL and release tag for the current package state, but no archival DOI has yet been minted for this specific example package.

A.7 Public provenance note

The journal supplement records that the public tables were identified with 1% PrSM FDR and 5% proteoform-level FDR in TopPIC and quantified across duplicate runs in TopDiff. The same supplement also embeds `Te12` and `Teo2` sheets that match the standalone PRIDE XLSX tables exactly. The PRIDE landing page lists the four raw duplicate files, but this package does not claim a raw-file reprocessing study.

A.8 Exact rebuild path

The package can be rebuilt locally with:

```
cd examples/codex-zebrafish-brain-paper-14rounds
./code/build_inputs.sh
./code/build_paper.sh
```

The analysis step should regenerate the summary metrics, exact-ID overlap files, marker-membership and robustness files, the run-pair and bootstrap sensitivity files, and the figure assets. The paper step should regenerate `paper/main.pdf`.

A.9 Marker-panel curation rule

The marker panel is intentionally conservative. A family enters the panel only if the source proteomics paper names it in the biological interpretation of telencephalon or optic tectum and if that interpretation can be connected to established zebrafish regional function in the cited background literature. The curated counts and family labels are materialized in `data/curated_evidence.json`, while the corresponding exact source-table members, row numbers, and matching rules appear in the marker-membership CSV and the traceability JSON.

A.10 Why the curation unit is still a marker family

The released region tables now do expose reusable machine-readable proteoform IDs, and this version of the package takes advantage of that fact. Even so, the strongest biological interpretation in the paper still runs through the source article's named marker families rather than through a full proteoform-by-proteoform atlas. That is deliberate. The exact-ID files strengthen traceability and missingness sensitivity, but the functional claim remains aligned with the families that the source paper itself foregrounded as biologically interpretable.

A.11 Leave-one-out and sensitivity files

The full leave-one-out table and the reviewer-driven sensitivity scenarios are saved as `results/leave_one_out_alignment.csv` and `results/sensitivity_scenarios.csv`. The main text and Figure 3 summarize the same robustness story.

A.12 Claim-to-evidence ladder

Table 6: Main claims, supporting evidence, and corresponding limits.

Claim	Evidence	Main limit
Proteoform-level regional separation	Article Jaccard 0.0434, exact-ID Jaccard 0.0360, occupancy-adjusted Jaccard 0.0432, 5% misidentification ceiling 0.0540, protein overlap 0.2151	Public tables, not raw-feature reprocessing
Functional markers align with expected region	Axis alignment, exact test, sign check, leave-one-out table, protein collapse, intensity weighting	Curated marker panel only
PTM layer contains a suggestive but unstable acetylation asymmetry	Unadjusted acetylation 14/23 versus 16/102; detectability and covariate sensitivity files	Public tables do not separate biology cleanly from geometry/coverage bias
Package improves auditability	Local code, figures, manifests, exact marker membership, and evidence traceability	Does not replace the source experiment

A.13 Claim-to-file map

The repository-level package keeps the full claim-to-file map in the local README and machine-readable manifests. In-paper, the important point is simply that every headline result in the main text is backed by a corresponding local CSV/JSON artifact plus the figure that summarizes it.